Lecture 21 Scribe

L21_L22.pdf
PDF

Generate a formal lecture scribe for Lecture 21 that serves as a high-fidelity, exam-ready reconstruction of the attached lecture slides . You must strictly adhere to the provided materials, ensuring that no content, intuition, or information is included unless it is explicitly traceable to the attached context. If a theorem or result is stated without a proof in the provided materials, you must not include or search for a proof; do not assume knowledge from previous lectures or external sources. The scribe must contain Definitions and notation, Assumptions and conditions, Statements of theorems or results, Proofs or proof sketches, step by step, Worked examples, with intermediate steps, Clear logical flow between ideas if present in the lecture slide.The scribe must be organized into a logical, hierarchical structure using sections and headings, documenting all definitions, notations, assumptions, and results exactly as presented. Every worked example must include the specific intermediate steps provided in the text. Your output must prioritize technical precision and strict faithfulness to the PDF, excluding any extra explanations, tutorials, or "common knowledge" not found within the document.

Below is a faithful scribe reconstructed from the uploaded combined slide deck **L21_L22.pdf**. The file is a **combined Lecture 21 and Lecture 22 deck**, and the uploaded material does **not visibly mark a clean boundary** between where Lecture 21 ends and Lecture 22 begins. So the scribe below is restricted to what is explicitly present in the provided PDF and follows its slide order and content only.

Lecture Scribe: Inequalities and Tail Bound

1. Title and Scope

Course: CSE400 - Fundamentals of Probability in Computing
Lecture Deck Title: Lecture 21 and L22: Inequalities and Tail Bound
Instructor: Dhaval Patel, PhD

Outline stated in the slides

- Tail and Motivation of Tail Bound
- Markov's Inequality
- Definition and Proof
- Example
- Chebyshive's Inequality
- Definition and Proof
- Example
- Chernof Bound and Poison Tail
- Definition and Proof
- Example
- Jensen's Inequality
- Convex Functions
- Proof and Example
- Case Study and System Level Understanding

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2. Tails and Motivation

2.1 Definition of Tail

Definition. The tail of a random variable X is

$$P\{X > x\}.$$

2.2 Why we care about tails

The slides list:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

2.3 Key idea

The slides state:

- Mean $\longrightarrow$ average behavior (easy)
- Tail $\longrightarrow$ rare extreme events (hard)

2.4 Why tails are hard to compute

The slide states:

- Requires summing many small probabilities

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3. Tail Examples

3.1 Example 1: Jobs assigned to servers

Question. Suppose you are distributing n jobs among n servers at random. What is the probability that a particular server gets $\geq k$ jobs?

Intuition from the slides

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This is identified on the slide as a **tail event** (rare overload).

Distribution and exact tail expression

$$X \sim \text{Binomial}(n, p)$$

and

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}.$$

3.2 Example 2: Poisson arrivals

Question. Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour. What is the probability that $\geq k$ jobs arrive during the first hour?

Intuition from the slides

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes, we see a burst of arrivals

This is identified on the slide as a **tail event** (sudden spike in load).

Distribution and exact tail expression

$$X \sim \text{Poisson}(\lambda)$$

and

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}.$$

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4. Why Tail Bounds?

The slides state:

- Exact tail probabilities are often **hard to compute**

Exact computation examples

$$P\{X \geq k\} = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$P\{X \geq k\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

with the note:

Complicated, long sums!

Bounding idea

Instead of computing the exact value, the slides propose:

- Find an **upper bound**
- Easier to compute
- Still gives a **guarantee**

Displayed idea:

$$\text{Exact: } P(X \geq k) = ? \qquad \text{Bound: } P(X \geq k) \leq \text{something simple}$$

Key idea: Approximate safely instead of computing exactly.

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5. Running Example Used Throughout

The slides introduce one common example to test multiple bounds.

Experiment

Flip a fair coin n times.

Random variable

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

Mean

$$E[X] = \frac{n}{2}$$

Tail question

$$P\left(X \geq \frac{3n}{4}\right)$$

The tail region is described as:

- getting unusually many heads

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6. Markov's Inequality

6.1 Key idea

The slides state:

- Uses only the **mean**
- Large values $\Rightarrow$ rare

Example note:

- Avg $= 10 \Rightarrow$ few values can be $100+$

6.2 Statement

Markov Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

This is presented for non-negative X.

6.3 Proof idea from the slides

The proof slide states:

- Consider only $X \geq a$
- Replace X by a

Then

$$\begin{aligned} E[X] &= \sum_{x=0}^{\infty} x\, p_X(x) \\ &\geq \sum_{x=a}^{\infty} x\, p_X(x) \\ &\geq \sum_{x=a}^{\infty} a\, p_X(x) \end{aligned}$$

$$= a \sum_{x=a}^{\infty} p_X(x)$$

$$= a \cdot P(X \geq a).$$

Therefore,

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

6.4 Markov on the running example

For the coin-flip example,

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right), \qquad \mu = E[X] = \frac{n}{2}.$$

We want

$$P\left(X \geq \frac{3n}{4}\right).$$

Applying Markov:

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{E[X]}{3n/4} = \frac{n/2}{3n/4} = \frac{2}{3}.$$

The slide explicitly comments:

Clearly a terrible bound because doesn't involve n.

6.5 Markov real-life example: income distribution

Scenario

- Average income $= 10{,}000$

Question

What fraction of people earn $50{,}000$?

Using Markov

$$P(X \geq 50{,}000) \leq \frac{10{,}000}{50{,}000} = 0.2.$$

Result from the slide

At most **20%** people can earn $50{,}000$.

6.6 Solved Example 1: Exam grades

Problem. The average grade on the first probability exam is 70%. The "A" grade cutoff is 90%. Find an upper bound on the fraction of students who receive an "A". Use Markov's inequality.

Solution

Step 1. Let $X \geq 0$ be student grade. Identify the mean.

$$\mu = E[X] = 70$$

Step 2. State Markov's inequality for non-negative X.

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 3. Substitute $\mu = 70$, threshold $a = 90$.

$$P(X \geq 90) \leq \frac{70}{90}$$

Step 4. Simplify the fraction.

$$P(X \geq 90) \leq \frac{7}{9}$$

Result

$$P(X \geq 90) \leq 0.778$$

So at most **77.8%** of students can receive an "A".

6.7 Solved Example 2: Server request load

Problem. A server processes an average of $5{,}000$ requests per second. Use Markov's inequality to find an upper bound on the probability that the server receives $20{,}000$ or more requests in a given second.

Solution

Step 1. Let X = requests/sec. Identify the mean.

$$\mu = E[X] = 5{,}000$$

Step 2. Define the overload threshold.

$$a = 20{,}000$$

Step 3. State Markov's inequality.

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Step 4. Substitute the known values.

$$P(X \geq 20{,}000) \leq \frac{5{,}000}{20{,}000}$$

Result

$$P(X \geq 20{,}000) \leq 4^{-1} = 0.25$$

The slide states: At most **25%** chance of $\geq 20{,}000$ requests.

6.8 Homework problem shown in the slides

Income distribution

In a specific city, the average annual income is $40{,}000$.

(a) Use Markov's inequality to find an upper bound on the probability that a randomly selected person earns $120{,}000$ or more.

(b) Can you use Markov's inequality to bound the probability that a person earns less than $10{,}000$? Why or why not?

Hints shown

- For (a): Define $X \geq 0$, state $E[X] = 40{,}000$, and apply $P(X \geq a) \leq E[X]/a$ with $a = 120{,}000$.
- For (b): Markov bounds only the right tail $P(X \geq a)$. Think about whether it applies to $P(X < a)$ directly, and what additional information would be required to handle the left tail.

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7. Chebyshev's Inequality

7.1 Key idea

The slides state:

- Uses mean + variance
- Controls deviation from mean

Markov is shown as

$$P(X \geq a) \leq \frac{E[X]}{a}$$

and Chebyshev is motivated as bounding spread around the mean.

7.2 Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}.$$

7.3 Proof idea from the slides

The slide says: apply Markov on $(X - \mu)^2$.

Then

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$

and by Markov,

$$P((X - \mu)^2 \geq a^2) \leq \frac{E[(X - \mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}.$$

Hence,

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}.$$

7.4 Chebyshev on the running example

For

$$X \sim \mathrm{Binomial}\left(n, \frac{1}{2}\right), \qquad \mu = \frac{n}{2}, \qquad \mathrm{Var}(X) = \frac{n}{4},$$

we want

$$P\left(X \geq \frac{3n}{4}\right).$$

The slide rewrites this as

$$P\left(X \geq \frac{3n}{4}\right) = P\left(X - \frac{n}{2} \geq \frac{n}{4}\right) \leq P\left(\left|X - \frac{n}{2}\right| \geq \frac{n}{4}\right).$$

Applying Chebyshev:

$$P\left(\left|X-\frac{n}{2}\right| \geq \frac{n}{4}\right) \leq \frac{n/4}{(n/4)^2} = \frac{4}{n}.$$

The slide then shows the one-sided bound as

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{1}{2} \cdot \frac{4}{n} = \frac{2}{n}.$$

Key insight stated on the slide

- Bound decreases as n increases
- Better than Markov (which was constant)

7.5 Real-life example: exam scores

Scenario

- Mean $= 70$ marks
- Std deviation $= 10$

Question

What fraction of students score below 40 or above 100?

These values are 30 away from the mean, so $a = 30$.

Using Chebyshev:

$$P(|X-70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}.$$

Result from the slide

At most $\frac{1}{9}$ students are far from mean.

7.6 Solved Example 1: Exam grades with variance

Problem. The average grade is 70% and the standard deviation is 5%. The "A" cutoff is 90%. Apply Chebyshev's inequality to obtain a tighter upper bound on the fraction of "A" students.

Solution

Step 1. Identify mean and variance.

$$\mu = 70, \qquad \sigma = 5, \qquad \sigma^2 = 25$$

Step 2. Note $X \geq 90$ implies $|X - 70| \geq 20$, so apply a one-sided bound through Chebyshev.

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

Step 3. State Chebyshev with $a = 20$.

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Step 4. Substitute $\sigma^2 = 25$, $a = 20$.

$$P(|X - 70| \geq 20) \leq \frac{25}{400}$$

Result

$$P(X \geq 90) \leq \frac{1}{16} = 0.0625$$

The slide notes this is much tighter than Markov's bound 0.778.

7.7 Solved Example 2: Network latency SLA

Problem. A network route has average latency $E[X] = 50$ ms and variance $\mathrm{Var}(X) = 16$ ms^2. Bound the probability that a packet's latency falls outside the range $[30 \text{ ms}, 70 \text{ ms}]$.

Solution

Step 1. Identify given parameters.

$$\mu = 50, \qquad \sigma^2 = 16$$

Step 2. Rewrite the failure condition as absolute deviation.

$$X \notin [30, 70] \iff |X - 50| \geq 20$$

Step 3. Apply Chebyshev with $a = 20$.

$$P(|X - 50| \geq 20) \leq \frac{16}{20^2} = \frac{16}{400}$$

Step 4. Simplify.

$$\frac{16}{400} = \frac{1}{25}$$

Result

$$P(|X-50| \geq 20) \leq 0.04$$

So at most **4%** of packets fall outside $[30, 70]$ ms.

7.8 Homework problem shown in the slides

Manufacturing tolerances

A factory produces resistors with average resistance $\mu = 100\,\Omega$ and standard deviation $\sigma = 2\,\Omega$. A resistor is defective if its resistance is outside $[94\,\Omega, 106\,\Omega]$.

(a) Use Chebyshev's inequality to find an upper bound on the probability that a resistor is defective.
(b) If $10{,}000$ resistors are produced, what is the maximum expected number of defective units?

Hints shown

- For (a): Express "defective" as $|X-\mu| \geq a$, where $a = 6\Omega$, and apply

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

- For (b): Multiply the probability bound from (a) by $10{,}000$.

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8. Chernoff Bound

8.1 Progression of ideas

The slides present the sequence:

- Markov: bound using X
- Chebyshev: bound using $(X-\mu)^2$
- Chernoff: bound using e^{tX}

8.2 Key insight

The slides state:

- Exponential grows very fast
- Large values of X become much larger
- Tail events get amplified

and:

Better transformation $\Rightarrow$ tighter tail bound

The slide intuition says:

- Small $X \to$ small e^{tX}
- Large $X \to$ very large e^{tX}

8.3 Core trick

Key transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

for $t > 0$.

Apply Markov to e^{tX}

Since e^{tX} is non-negative,

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}.$$

Thus,

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}}.$$

The slide emphasizes:

- e^{tX} is always non-negative
- So Markov inequality applies
- Now we have a bound depending on t

8.4 Final upper-tail Chernoff form

The slides then minimize over $t > 0$:

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}.$$

Key insight on the slide

- Bound holds for all $t > 0$
- Choose best $t \Rightarrow$ tightest bound

The slide comments:

- Much stronger than Markov + Chebyshev
- Exponential decay instead of polynomial

8.5 Lower-tail form

For the lower tail, the slides state the goal:

$$\text{Bound } P(X \le a)$$

and use $t < 0$.

Trick

$$P(X \le a) = P(e^{tX} \ge e^{ta}), \qquad t < 0$$

leading to

$$P(X \le a) \le \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}.$$

The slide notes:

- Same idea, different direction (left tail)

8.6 Chernoff bound for Poisson tail

The slides specialize Chernoff to $X \sim \text{Poisson}(\lambda)$.

Goal

Bound tail of $X \sim \text{Poisson}(\lambda)$.

Step 1: derive $E[e^{tX}]$ for $t > 0$

Starting from the Poisson pmf,

$$\begin{aligned} E[e^{tX}] &= \sum_{i=0}^{\infty} e^{ti} \cdot \frac{e^{-\lambda}\lambda^i}{i!} \\ &= e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda e^t)^i}{i!} \\ &= e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda(e^t - 1)}. \end{aligned}$$

Step 2: apply Chernoff for $a > \lambda$

$$P(X \ge a) \le \min_{t>0} \left\{ e^{\lambda(e^t-1)-ta} \right\}.$$

To minimize the bound, the slide states:

- It suffices to minimize the exponent
- Exponent is minimized at

$$t = \ln(a/\lambda).$$

Substituting this value yields the final bound shown on the slide:

$$P(X \geq a) \leq e^{-\lambda} \left(\frac{e\lambda}{a}\right)^a.$$

8.7 Solved Example 1: Binomial lower tail (job interviews)

Problem. Jiacheng has independent probability $p = 0.5$ of being hired at each of $n = 20$ companies. Using $P(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}$, bound the probability he receives zero job offers.

Solution

Step 1. Define X and compute $\mu = np$.

$$X \sim \text{Binomial}(20, 0.5), \qquad \mu = np = 10$$

Step 2. Set $(1-\delta)\mu = 0$ and solve for δ.

$$0 = (1-\delta)(10) \implies \delta = 1$$

Step 3. State the Chernoff lower-tail bound.

$$P(X \leq 0) \leq \exp\left(-\frac{\mu\delta^2}{2}\right)$$

Step 4. Substitute $\mu = 10$, $\delta = 1$.

$$P(X \leq 0) \leq \exp\left(-\frac{10}{2}\right) = e^{-5}$$

Result

$$P(X \leq 0) \leq e^{-5} \approx 0.0067$$

The slide states: Less than **0.67%** chance of receiving zero offers.

8.8 Solved Example 2: Deriving the lower-tail Chernoff bound

Problem. Derive the general structure of the Chernoff lower-tail bound, i.e. an upper bound on $P(X \leq a)$.

Solution from the slides

Step 1. Choose $t < 0$. Multiplying by negative flips the inequality.

$$P(X \leq a) = P(tX \geq ta), \qquad t < 0$$

Step 2. Exponentiate both sides (e^x is strictly increasing).

$$P(tX \geq ta) = P(e^{tX} \geq e^{ta})$$

Step 3. Apply Markov's inequality.

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Step 4. Minimize over $t < 0$.

$$P(X \leq a) \leq e^{-ta} E[e^{tX}]$$

and finally

$$P(X \leq a) \leq \min_{t<0} \left\{ e^{-ta} E[e^{tX}] \right\}.$$

Result

Tightest bound optimizes over $t < 0$.

8.9 Homework problem shown in the slides

Let Y be a random variable with $0 \leq Y \leq 1$ and $E[Y] = \mu$. Use the convexity of e^{tY} to prove:

$$E[e^{tY}] \leq 1 - \mu + \mu e^t.$$

Hints shown

Step 1. Since $Y \in [0, 1]$, write

$$Y = (1 - Y) \cdot 0 + Y \cdot 1.$$

Step 2. Apply convexity of e^x:

$$e^{tY} \leq (1 - Y)e^0 + Ye^t = 1 - Y + Ye^t.$$

Step 3. Take expectations:

$$E[e^{tY}] \leq 1 - E[Y] + E[Y]e^t = 1 - \mu + \mu e^t.$$

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9. Jensen's Inequality

9.1 Initial motivation

The slides begin with the known fact

$$E[X^2] \geq E[X]^2$$

because variance is non-negative:

$$\mathrm{Var}(X) = E[X^2] - E[X]^2 \geq 0.$$

Then the slides ask whether similarly:

$$E[X^3] \geq E[X]^3, \qquad E[X^4] \geq E[X]^4, \qquad E[X^5] \geq E[X]^5 \ ?$$

The slide then states the result:

For any positive random variable X, for all $a \in \mathbb{R}$,

$$E[X^a] \geq E[X]^a$$

and says this is a consequence of Jensen's inequality.

9.2 Convexity definition

Definition. Convex Function.
A function $g(x)$ is convex on interval S if, for any $x_1, x_2 \in S$ and $\alpha \in [0, 1]$,

$$g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha) g(x_2).$$

Plot interpretation shown on the slides

- Pick two points x_1 and x_2
- Blue curve = actual function $g(x)$
- Curve always stays below the line

The slide also states:

$$g(x) \text{ is convex on } S \iff g''(x) \geq 0.$$

9.3 From convexity to expectation

The slides first present convex combination form:

If $\sum_{i=1}^{n} \alpha_i = 1$, then

$$g\left(\sum_{i=1}^{n}\alpha_i x_i\right) \leq \sum_{i=1}^{n}\alpha_i g(x_i).$$

Interpretation given on the slides

- α_i act like probabilities
- $\sum \alpha_i = 1$
- This represents a weighted average

Probability interpretation

- $X = x_i$ with probability $p_X(x_i)$
- Expectation = weighted average

Hence mathematically,

$$E[X] = \sum p_X(x_i)x_i$$

$$E[g(X)] = \sum p_X(x_i)g(x_i)$$

and applying convexity,

$$g(E[X]) \leq E[g(X)].$$

9.4 Jensen's inequality statement

Theorem (Jensen's Inequality).
If $g(x)$ is convex on an interval S and X takes values in S, then

$$g(E[X]) \leq E[g(X)].$$

Interpretation from the slides

- Average first $\rightarrow$ smaller value
- Apply function first $\rightarrow$ larger value

The slide concludes:

Convex functions push expectation upwards.

9.5 Simple example

Let X be a random variable with

$$X = \begin{cases} 1 & \text{with probability } 0.5 \\ 3 & \text{with probability } 0.5 \end{cases}$$

and take $g(x) = x^2$.

Step 1: expectation

$$E[X] = 0.5(1) + 0.5(3) = 2$$

Apply function after averaging:

$$g(E[X]) = (2)^2 = 4$$

Step 2: apply function first

$$E[g(X)] = 0.5(1)^2 + 0.5(3)^2 = 5$$

Hence

$$g(E[X]) = 4 \leq 5 = E[g(X)].$$

9.6 Solved Example 1: Verifying Jensen numerically

Problem. Let $X = 2$ with probability 0.5 and $X = 4$ with probability 0.5. Let $g(x) = x^2$ (convex). Verify numerically that $g(E[X]) \leq E[g(X)]$.

Solution

Step 1. Compute the expectation of X.

$$E[X] = (2)(0.5) + (4)(0.5) = 3$$

Step 2. Compute left side of Jensen: apply g to the mean.

$$g(E[X]) = (3)^2 = 9$$

Step 3. Find the distribution of $g(X) = X^2$.

$$P(X^2 = 4) = 0.5, \qquad P(X^2 = 16) = 0.5$$

Step 4. Compute right side of Jensen.

$$E[g(X)] = (4)(0.5) + (16)(0.5) = 10$$

Result

$$g(E[X]) = 9 \leq 10 = E[g(X)].$$

The slide notes:

- Strict inequality holds
- g is convex

- X is non-constant

9.7 Homework problem shown in the slides

Let $X > 0$ be a random variable representing server processing time with $E[X] = \mu$. The cost function is $C(X) = X^2$. Use Jensen's inequality to prove that

$$E[X^2] \geq (E[X])^2.$$

Hints shown

Step 1. Set $g(x) = x^2$.
Step 2. Prove convexity: $g''(x) = 2 \geq 0$, so g is convex.
Step 3. Apply Jensen:

$$g(E[X]) \leq E[g(X)]$$

which gives

$$(E[X])^2 \leq E[X^2].$$

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10. Case Study and System-Level Understanding

10.1 Case study question

Case Study: How to predict the worst case demands on server?

The slide includes real-world examples of demand spikes and service failures.

10.2 Step 1: Real-world system failures

The slides ask us to imagine an e-commerce sale.

Given points

- Millions of users log in at midnight
- Average traffic is manageable
- Sudden, extreme traffic spikes cause server queues to overflow
- The system crashes. Revenue is lost.

Engineering problem

We cannot design hardware for "average" load. We must design for the **tail** (extreme spikes).

10.3 Step 2: Mapping demand to probability

Let:

- **Demand** X: incoming requests per second (a random variable)
- **Capacity** C: maximum requests the server cluster can handle
- **Failure event**:

$$X \geq C$$

(demand exceeds capacity)

SLA stated in the slides

Management demands 99% uptime.

Probabilistic rule:

$$P(X \geq C) \leq 0.01.$$

Question: How large must we build C to guarantee this?

10.4 Step 3a: Applying Markov's bound

Scenario

We only know the average traffic.

Telemetry shows average load

$$\mu = 10{,}000 \text{ req/sec}.$$

Need

$$P(X \geq C) \leq 0.01.$$

Math

Using Markov,

$$P(X \geq C) \leq \frac{\mu}{C}.$$

Set

$$\frac{10{,}000}{C} = 0.01$$

which gives

$$C = 1{,}000{,}000 \text{ req/sec}.$$

System inference #1

If you only know the mean, the math forces you to buy **100x more hardware** to be safe.

10.5 Step 3b: Applying Chebyshev's bound

Scenario

We know the average and the burstiness (variance).

Given:

- Mean $\mu = 10{,}000$ req/sec
- Standard deviation $\sigma = 2{,}000$ req/sec

Still require

$$P(X \geq C) \leq 0.01.$$

Math

Chebyshev:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Let

$$a = C - \mu.$$

Then solve

$$\frac{2000^2}{(C - 10000)^2} = 0.01.$$

The slides give:

$$C = 30{,}000 \text{ req/sec}.$$

System inference #2

We reduced capacity from $1{,}000{,}000$ to $30{,}000$.

10.6 Step 3c: Applying the Chernoff bound

Scenario

We know the exact traffic distribution pattern.

The slides state:

- We know traffic follows a Poisson distribution
- Chernoff uses the entire Moment Generating Function
- It bounds the tail exponentially:

$$P(X \geq C) \leq e^{-I(C)}.$$

System result

Solve

$$e^{-I(C)} \leq 0.01.$$

The slides state:

$$C \approx 14{,}500 \text{ req/sec}.$$

System inference #3

Chernoff gives precise scaling with minimal hardware.

10.7 Step 4: Final system design inferences

The slides give the following comparison.

Inequality	Telemetry Available	Required Capacity	Use Case
Markov	Only Average Load	1,000,000	Unknown traffic (Safety fallback)
Chebyshev	Avg + Burstiness	30,000	Google SRE general provisioning
Chernoff	Full Distribution	14,500	Netflix/AWS exact node sizing

Final rule stated on the last slide

The Golden Rule of Probability in Systems:

More statistical information $\Rightarrow$ tighter bounds $\Rightarrow$ lower hardware costs.

If you want, I can now turn this into **same-to-same LaTeX scribe code** in a clean exam-ready format.